



RV College of Engineering

(An Autonomous Institution Affiliated to VTU)

R V Vidyanikethan Post
Mysuru Road, Bengaluru - 560059

VI Semester B.E. Regular / Supplementary Examination July / August 2026

(Common to CS/IS)

Web Frameworks

Time : 3 Hours

Instructions to the students

Maximum Marks : 100

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer Five full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

Part A

Question No	Question	M	CO	BT
1.1	Find the output of the following JavaScript code. <pre><body><script type="text/javascript"> function display() { var c= 50+20+"welcome"; var d="welcome"+50+20; document.write(c); document.write(d); } display(); </script></body></pre>	02	2	1
1.2	What is the output of the following code after performing slice operation in array elements? <pre><script type="text/javascript"> var a=new Array(34,67,78,90); var b=a.slice(2,4); document.write(b); </script></pre>	02	1	3

Find the output of the following JavaScript code.

```
<script type="text/javascript">
```

```
var str1="programming lang";
```

```
var i2=str1.search(/prog$/);
```

1.3 if(i2==0)

```
document.write("Found");
```

```
else
```

```
document.write("NotFound");
```

```
</script>
```

Write the output of the following JavaScript code.

```
<script type="text/javascript">
```

```
var st="yykkkkk";
```

1.4 var i=st.match(/y*z+k?/);

```
document.write(i);
```

```
</script>
```

Find the output of the following JavaScript code.

```
<script type="text/javascript">
```

```
var str1="scripting language";
```

```
var i2=str1.search(/^scri/);
```

1.5 if(!i2)

```
document.write("Found");
```

```
else
```

```
document.write("NotFound");
```

```
</script>
```

1.6 List out all the event types that are not allowed to bubble in DOM2 model event propagation.

1.7 Which method is used to access elements by class name in JavaScript?

1.8 While creating an XML document, if the value for an attribute for an element is optional, then which value should be used while defining an attribute in Document Type Definition?

Consider the below DTD:

1.9 <!ENTITY writer "Donald.">

```
<!ENTITY copyright "Copyright_XML_Xhtml">
```

Write the XML code that shows the correct way of accessing the entity data.

1.10 Name any two AngularJS event directives.

1.11 What command is used to start the React local development server?

1.12	What is the role of require() in Node.js?	01	4	2
1.13	What is the significance of HTTP status code 200 in AJAX?	01	2	2
1.14	Name any two popular applications that use AJAX.	02	3	2
1.15	What is the purpose of the manage.py file in Django?	01	2	2

Part B

Question No	Question	M	CO	BT
2a	Write a JavaScript function to check whether the given Email-ID is valid or not.	08	2	4
2b	Write a JavaScript program to sort the items of an array in an ascending order and descending order	08	3	3
3a	Briefly discuss the concept of DOM(Document Object Model) in JavaScript.	08	1	2
3b	Create the XHTML document that must contain three paragraphs of text, stacked on top of each other, with only enough of each showing so that the mouse cursor can always be placed over some part of them. When the cursor is placed over the exposed part of any paragraph, it should rise to the top to become completely visible.	08	2	4
OR				
4a	Develop a XHTML document that has checkboxes for 3 different fruits along with a submit button. Each of the checkboxes should have its own "onclick" event handler. These handlers must add the cost of the fruits to a total cost and the event handler for the submit button must produce an alert window with the total cost as a message.	08	3	6
4b	Describe all of the differences between the three possible values of the position property.	08	2	1
5a	i) Create a Document Type Definition(DTD) for a catalog of cars, where each car has the child elements make, model, year, color, engine, number_of_doors, transmission_type and accessories. The engine element has the child elements, number_of_cylinders and fuel_system. The accessories element has attribute such as radio, air_conditioning, power_windows, power_steering and poer_brakes, each of which is required and has the possible values yes and no. Entities must be declared for the names of popular car makes. ii) Create an XML document with at least three instances of the car element defined in the above DTD and Process this document using the DTD and produce a display of raw XML document. iii) Create an XSLT style sheet for one car element of the XML document and use it to create a display of that element. Write a PHP script to sort the following associative array : array("Sophia"=>"31","Jacob"=>"41","William"=>"39","Ramesh"=>"40") in	08	4	6
5b	i) ascending order sort by Value ii) ascending order sort by Key iii) descending order sorting by Value iv) descending order sorting by Key	08	3	3

OR

6a	Write a PHP program to calculate the total, percentage, and grade of a student based on marks entered through an HTML form.	08	4	4
6b	Consider the catalog of cars, where each car has the child elements make, model, year, color, engine, number_of_doors, transmission_type and accessories. The engine element has the child elements, number_of_cylinders and fuel_system. The accessories element has attribute such as radio, air_conditioning, power_windows, power_steering and poer_brakes, each of which is required and has the possible values yes and no. Entities must be declared for the names of popular car makes. For the above data, i) Create a XML Schema for the XML document described above. ii) Create a CSS style sheet for the XML document and use it to create a display of that document.	08	3	4
7a	Write short notes on Angular JS Modules.	08	4	2
7b	How the event handling works in React.js? Illustrate it with an example.	08	4	3
OR				
8a	Define AngularJS expressions. Illustrate how expressions can be used with arithmetic operations, strings, objects, and arrays with examples.	08	4	6
8b	Explain the Event Loop architecture and its advantages in Node JS.	08	4	1
9a	Discuss the role of JavaScript, DOM, and XMLHttpRequest in AJAX-based applications.	08	4	1
9b	Explain the advantages of Django over traditional web development frameworks.	08	4	2
OR				
10a	Describe the lifecycle of an XMLHttpRequest object with suitable examples.	08	4	3
10b	Describe Django's Validation mechanism. Explain the difference between: (i) Built-in validation —validators like max_length, required, EmailField; and (ii) Custom validation — using clean_<fieldname>() method and raising ValidationError. Write a form that validates a username contains only alphabets.	08	4	3